

Benjamin T. James

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Cambridge, MA 02139, USA

Education

- **Massachusetts Institute of Technology** September 2019 - May 2026
Ph.D in Electrical Engineering and Computer Science, minor in Econometrics Cambridge, Massachusetts
 - GPA: 4.80/5.00
 - Advisor: Manolis Kellis
- **Massachusetts Institute of Technology** September 2019 - May 2023
SM in Electrical Engineering and Computer Science Cambridge, Massachusetts
 - GPA: 4.80/5.00
 - Advisor: Manolis Kellis
- **University of Tulsa** Aug 2015 - May 2019
BSc in Mathematics, BSc in Computer Science Tulsa, Oklahoma
 - GPA: 3.747/4.000
 - Advisor: Hani Z. Girgis

Projects

- **Single-cell dissection of the human brain** 2021 - 2026
Tools: R, Python, Bash [🔗]
 - Expert level analysis of several large-sample snRNA-seq and snATAC-seq atlases and application towards human disease (Alzheimer's disease, addiction), including several Cell publications (two co-first authorship [S.1], [J.11])
 - Revealed differentially expressed genes and differentially accessible regions, prioritized genetic variants, and proposed downstream pathways dysregulated in disease
 - Developed peak-gene linking, snATAC-seq/snRNA-seq integration methodology, and gene/enhancer module discovery in service of understanding gene regulatory changes in human disease [J.11]
 - Led computational analysis for a single-cell dissection of the human striatum [S.1], transcriptomic rescue of mouse chemobrain symptoms [J.6], and organoid technology [J.5] but played integral role in others
 - Wrote several bespoke scripts across several single-cell projects and organized into [this repo](#) that much of the lab uses to perform analysis at the scale of hundreds of samples
- **Single cell genomics module discovery for grouping functional units** 2023-2026
Tools: Python, R, C++ [🔗]
 - Developed gene co-expression framework (similar to WGCNA) to categorize differential gene expression across multiple brain regions in Alzheimer's disease [J.7], [J.11]
 - Utilizes PCA loadings from high dimensional genomic assays to relate gene embeddings to functional graphs while accounting for technical artifacts like sequencing depth
 - Although mature, I am trying to still improve estimation (Q-Q plots, modelling technical artifacts in-model instead of after-model, etc) using high-dimensional statistics and linear algebra
 - Co-developed with Carles Boix, used prominently in [J.7] and [J.11] to interpret differentially expressed genes across multiple cell types and brain regions
- **Scalable, spatially-informed clustering for spatial transcriptomics** 2025-2026
Tools: Python, ARPACK (scipy) [🔗]
 - Adaptation of the [BANKSY algorithm](#) for high dimensional transcriptomics

- Newer spatial transcriptomics assays like 10x Genomics Visium HD contain millions of spots per experiment that are not amenable to these methods
- Tech specific adaptations include a Sinkhorn-Knopp based noise removal (de-stripping) and a Laplacian of Gaussian filter (for blob detection) replacing a Gabor filter
- Sparse incremental Lanczos-based PCA that allows for input matrices to stay sparse, as opposed to the sklearn based IncrementalPCA
- Designed and implemented alone, with no collaborators and no advisor input

• Linking regulatory elements to their target genes

2019-2023

Tools: Python, R, XGBoost

[🔗]

- Developed a method that links annotated enhancers to target genes by using correlated activity between epigenomes (imputed ChIP-seq) over putative enhancers and corresponding gene expression [J.12], [J.9], [P.18], [J.2]
- Started as rotation project in Manolis Kellis lab, directly supervised under Carles Boix (1st author of Nature paper), where my contributions placed me as a second author on a Nature paper [J.12]
- Worked with the ENCODE Distal Regulation group and GENCODE with this method to annotate regulatory elements [J.8], [J.4], [P.19]
- Later this methodology for linking putative enhancers to genes was extended for single-nucleus ATAC-seq studies for Alzheimer's disease [J.11], [J.10]

• Intelligent clustering of DNA sequences

2016-2019

Tools: C++/OpenMP; gdb/valgrind

[🔗]

- Developed a fast method as an undergraduate student to approximate an expensive method for DNA similarity (alignment). [J.17], [J.16], [J.14], [J.13], [P.20]
- Implemented a fast, concurrent version of the Mean Shift algorithm that scales well for large DNA sets in a manner similar to (H)DBSCAN
- Self-supervised linear model generates mutations at a known rate and then assigns their similarity

Publications

C=Conference, J=Journal, P=Preprint, S=In Submission, T=Thesis, *=Co-first authorship

- [S.1] Linville*, R. M., **James*, B. T.**, Galani, K., Ho, L., Shin, J., Oliver, E., Bock, R., Murray, E. M., Louca, M., Oke, O., Wang, C., Engelberg-Cook, E., DeTure, M., Farrell, V., Fitzwalter, B., Pineda, S., Sathitloetsakun, S., Liang, X., Madras, B., and Dickson, D. W., Mash, D. C., Turecki, G., Wheeler, V. C., Alvarez, V. A., Gabuzda, D., Kellis, M., Heiman, M. (2025). Cross-species single cell atlas of the striatum defines cell type and subregion disease vulnerabilities. *Cell*. **Accepted for publication pending minor revisions**
- [J.2] Boix, C. A., **James, B. T.**, Park, Y. P., Meuleman, W., & Kellis, M. (2025). Author Correction: Regulatory genomic circuitry of human disease loci by integrative epigenomics. *Nature*, 643(8071), E11. <https://doi.org/10.1038/s41586-025-09134-4>
- [J.3] Liu, Z., Zhang, S., **James, B. T.**, Galani, K., Mangan, R. J., Fass, S. B., Liang, C., Wagle, M. M., Boix, C. A., Tanigawa, Y., Yun, S., Sung, Y., Xiong, X., Sun, N., Hou, L., Wohlwend, M., Qiu, M., Han, X., Xiong, L., ... Kellis, M. (2025). Single-cell multiregion epigenomic rewiring in Alzheimer's disease progression and cognitive resilience. *Cell*, 188(18), 4980-5002.e29. <https://doi.org/10.1016/j.cell.2025.06.031>
- [J.4] Mudge, J. M., Carbonell-Sala, S., Diekhans, M., Martinez, J. G., Hunt, T., Jungreis, I., Loveland, J. E., Arnan, C., Barnes, I., Bennett, R., Berry, A., Bignell, A., Cerdán-Vélez, D., Cochran, K., Cortés, L. T., Davidson, C., Donaldson, S., Dursun, C., Fatima, R., ... Frankish, A. (2025). GENCODE 2025: reference gene annotation for human and mouse. *Nucleic Acids Research*, 53(D1), D966-D975. <https://doi.org/10.1093/nar/gkae1078>
- [J.5] Stanton, A. E., Bubnys, A., Agbas, E., **James, B.**, Park, D. S., Jiang, A., Pinals, R. L., Liu, L., Truong, N., Loon, A., Staab, C., Cerit, O., Wen, H.-L., Mankus, D., Bisher, M. E., Lytton-Jean, A. K. R., Kellis, M., Blanchard, J. W., Langer, R., & Tsai, L.-H. (2025). Engineered 3D immuno-glia-neurovascular human miBrain model. *Proceedings of the National Academy of Sciences*, 122(42), e2511596122. <https://doi.org/10.1073/pnas.2511596122>

- [J.6] Kim, T., **James, B. T.**, Kahn, M. C., Blanco-Duque, C., Abdurrob, F., Islam, M. R., Lavoie, N. S., Kellis, M., & Tsai, L.-H. (2024). Gamma entrainment using audiovisual stimuli alleviates chemobrain pathology and cognitive impairment induced by chemotherapy in mice. *Science Translational Medicine*, 16(737), eadf4601. <https://doi.org/10.1126/scitranslmed.adf4601>
- [J.7] Mathys, H., Boix, C. A., Akay, L. A., Xia, Z., Davila-Velderrain, J., Ng, A. P., Jiang, X., Abdelhady, G., Galani, K., Mantero, J., Band, N., **James, B. T.**, Babu, S., Galiana-Melendez, F., Louderback, K., Prokopenko, D., Tanzi, R. E., Bennett, D. A., Tsai, L.-H., & Kellis, M. (2024). Single-cell multiregion dissection of Alzheimer's disease. *Nature*, 632(8026), 858-868. <https://doi.org/10.1038/s41586-024-07606-7>
- [J.8] Frankish, A., Carbonell-Sala, S., Diekhans, M., Jungreis, I., Loveland, J. E., Mudge, J. M., Sisu, C., Wright, J. C., Arnan, C., Barnes, I., Banerjee, A., Bennett, R., Berry, A., Bignell, A., Boix, C., Calvet, F., Cerdán-Vélez, D., Cunningham, F., Davidson, C., ... Flicek, P. (2023). GENCODE: reference annotation for the human and mouse genomes in 2023. *Nucleic Acids Research*, 51(D1), D942-D949. <https://doi.org/10.1093/nar/gkac1071>
- [J.9] Hou, L., Xiong, X., Park, Y., Boix, C., **James, B.**, Sun, N., He, L., Patel, A., Zhang, Z., Molinie, B., Van Wittenberghe, N., Steelman, S., Nusbaum, C., Aguet, F., Ardlie, K. G., & Kellis, M. (2023). Multitissue H3K27ac profiling of GTEx samples links epigenomic variation to disease. *Nature Genetics*, 55(10), 1665-1676. <https://doi.org/10.1038/s41588-023-01509-5>
- [J.10] Sun, N., Victor, M. B., Park, Y. P., Xiong, X., Scannail, A. N., Leary, N., Prosper, S., Viswanathan, S., Luna, X., Boix, C. A., **James, B. T.**, Tanigawa, Y., Galani, K., Mathys, H., Jiang, X., Ng, A. P., Bennett, D. A., Tsai, L.-H., & Kellis, M. (2023). Human microglial state dynamics in Alzheimer's disease progression. *Cell*, 186(20), 4386-4403.e29. <https://doi.org/10.1016/j.cell.2023.08.037>
- [J.11] Xiong*, X., **James***, **B. T.**, Boix*, C. A., Park, Y. P., Galani, K., Victor, M. B., Sun, N., Hou, L., Ho, L.-L., Mantero, J., Scannail, A. N., Dileep, V., Dong, W., Mathys, H., Bennett, D. A., Tsai, L.-H., & Kellis, M. (2023). Epigenomic dissection of Alzheimer's disease pinpoints causal variants and reveals epigenome erosion. *Cell*, 186(20), 4422-4437.e21. <https://doi.org/10.1016/j.cell.2023.08.040>
- [J.12] Boix, C. A., **James, B. T.**, Park, Y. P., Meuleman, W., & Kellis, M. (2021). Regulatory genomic circuitry of human disease loci by integrative epigenomics. *Nature*, 590(7845), 300-307. <https://doi.org/10.1038/s41586-020-03145-z>
- [J.13] Girgis, H. Z., **James, B. T.**, & Luczak, B. B. (2021). Identity: rapid alignment-free prediction of sequence alignment identity scores using self-supervised general linear models. *Nar Genomics and Bioinformatics*, 3(1), lqab001. <https://doi.org/10.1093/nargab/lqab001>
- [J.14] Velasco, A. II, **James, B. T.**, Wells, V. D., & Girgis, H. Z. (2020). Look4TRs: a de novo tool for detecting simple tandem repeats using self-supervised hidden Markov models. *Bioinformatics*, 36(2), 380-387. <https://doi.org/10.1093/bioinformatics/btz551>
- [J.15] Luczak, B. B., **James, B. T.**, & Girgis, H. Z. (2019). A survey and evaluations of histogram-based statistics in alignment-free sequence comparison. *Briefings in Bioinformatics*, 20(4), 1222-1237. <https://doi.org/10.1093/bib/bbx161>
- [J.16] Zielezinski, A., Girgis, H. Z., Bernard, G., Leimeister, C.-A., Tang, K., Dencker, T., Lau, A. K., Röhlings, S., Choi, J. J., Waterman, M. S., Comin, M., Kim, S.-H., Vinga, S., Almeida, J. S., Chan, C. X., **James, B. T.**, Sun, F., Morgenstern, B., & Karlowski, W. M. (2019). Benchmarking of alignment-free sequence comparison methods. *Genome Biology*, 20(1), 144. <https://doi.org/10.1186/s13059-019-1755-7>
- [J.17] **James, B. T.**, Luczak, B. B., & Girgis, H. Z. (2018). MeShClust: an intelligent tool for clustering DNA sequences. *Nucleic Acids Research*, 46(14), e83. <https://doi.org/10.1093/nar/gky315>
- [P.18] Middelbeek, R. J. W., Yang, J., Nigro, P., **James, B.**, Papadopoulos, D., Simpson, L. K., Park, J., Vamvini, M., Ho, L.-L., Zhang, H., Carbone, N. P., Hirshman, M. F., Kellis, M., & Goodyear, L. J. (2025). Type 2 Diabetes and Obesity Alter Exercise Training-Induced Transcriptional Adaptations to Subcutaneous White Adipose Tissue. bioRxiv. <https://doi.org/10.1101/2025.10.11.681273>
- [P.19] Gschwind, A. R., Mualim, K. S., Karbalayghareh, A., Sheth, M. U., Dey, K. K., Jagoda, E., Nurtdinov, R. N., Xi, W., Tan, A. S., Jones, H., Ma, X. R., Yao, D., Nasser, J., Avsec, Ž., **James, B. T.**, Shamim, M. S., Durand, N. C., Rao, S. S. P., Mahajan, R., ... Engreitz, J. M. (2023). An encyclopedia of enhancer-gene regulatory interactions in the human genome. bioRxiv. <https://doi.org/10.1101/2023.11.09.563812> **Soon to be in Nature, pending rest of package approval**

- [P.20]** James, B. T., & Girgis, H. Z. (2018). *MeShClust2: Application of alignment-free identity scores in clustering long DNA sequences*. bioRxiv. <https://doi.org/10.1101/451278>
- [T.21]** James, B. T. (2023). *Correlation-based Linking of Epigenomic Regions to Target Genes in Bulk and Single Cell Epigenomic Data* [Thesis, Massachusetts Institute of Technology]. <https://dspace.mit.edu/handle/1721.1/151538>

Skills

- **Programming Languages:** R and Python (proficient), C++, Bash, C, AWK, Julia
- **Research-specific Packages and Tools:** Bioconductor, scverse, samtools, bedtools, IGV, Eigen, NumPy/SciPy/SK-Learn, PyTorch, brms
- **Operating Systems/DevOps:** De-facto manager of two lab-wide compute clusters, managing software (Lmod, conda, singularity environments, CUDA drivers etc). Experienced Linux user (Debian/Gentoo user since 2014), see [my dotfiles](#), formerly ArchLinux AUR [maintainer](#)

Honors and Awards

- **NSF GRFP** May 2019
National Science Foundation
 - Awarded NSF GRFP during undergraduate tenure
- **Cronin Fellowship** April 2019
MIT EECS department
 - MIT departmental (EECS) fellowship for exceptional Ph.D applicants